$$
\begin{aligned}
m\,m_1 = -1 \quad \text{or} \quad m\,m_2 &= -1 \\
\Rightarrow \quad (m\,m_1 + 1) = 0 \ \text{or} \ (m\,m_2 + 1) &= 0 \\
\Rightarrow \quad (m\,m_1 + 1)(m\,m_2 + 1) &= 0 \\
\Rightarrow \quad (m_1 m_2)m^2 + (m_1 + m_2)m + 1 &= 0 \\
\Rightarrow \quad \left(\frac{a}{b}\right)\left(-\frac{p}{q}\right)^2 + \left(-\frac{2h}{b}\right)\left(-\frac{p}{q}\right) + 1 &= 0 \\
\Rightarrow \quad ap^2 + 2hpq + bq^2 &= 0
\end{aligned}
$$

The pair of straight lines through the origin is a homogeneous equation of degree two

Example 6.41 Show that the straight lines joining the origin to the points of intersection of $3x - 2y + 2 = 0$ and $3x^2 + 5xy - 2y^2 + 4x + 5y = 0$ are at right angles

Solution:

The straight lines joining the origin and the points of intersection of given equations is a second degree homogeneous equation.

Following steps show, the way of homogenizing the $3x^2 + 5xy - 2y^2 + 4x + 5y = 0$ with $3x - 2y + 2 = 0$

$$
\begin{aligned}
3x^2 + 5xy - 2y^2 + (4x + 5y)(1) &= 0 \ \text{and} \ \frac{(3x - 2y)}{-2} = 1 \\
3x^2 + 5xy - 2y^2 + (4x + 5y)\left(\frac{3x - 2y}{-2}\right) &= 0 \\
(-2)(3x^2 + 5xy - 2y^2) + (4x + 5y)(3x - 2y) &= 0
\end{aligned}
$$

On simplification,

We get, $2x^2 - xy - 2y^2 = 0$. $\Rightarrow\ a = 2,\ b = -2 \ \Rightarrow\ a + b = 0$

Since sum of the co-efficient of x^2 and y^2 is equal to zero, the lines are at right angles

Exercise - 6.4

1. Find the combined equation of the straight lines whose separate equations are $x - 2y - 3 = 0$ and $x + y + 5 = 0$.
2. Show that $4x^2 + 4xy + y^2 - 6x - 3y - 4 = 0$ represents a pair of parallel lines.
3. Show that $2x^2 + 3xy - 2y^2 + 3x + y + 1 = 0$ represents a pair of perpendicular lines.
4. Show that the equation $2x^2 - xy - 3y^2 - 6x + 19y - 20 = 0$ represents a pair of intersecting lines. Show further that the angle between them is $\tan^{-1}(5)$.
5. Prove that the equation to the straight lines through the origin, each of which makes an angle α with the straight line $y = x$ is $x^2 - 2xy\ \sec 2\alpha + y^2 = 0$
6. Find the equation of the pair of straight lines passing through the point $(1, 3)$ and perpendicular to the lines $2x - 3y + 1 = 0$ and $5x + y - 3 = 0$